

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Science

Master of Science Programme
(Biotechnology/Microbiology/Biochemistry)

2022

Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy, Nursing and Paramedical Sciences, is one of the choicest destinations by students. It has been transformed into **Charotar University of Science and Technology (CHARUSAT)** through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC- Govt. of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs.3 Crores (INR 30 million). At present there are seven different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D. degrees. These faculties, in all offer 60+ different programmes. A quick glimpse is as under:

Faculty	Institute	Programmes Offered
Faculty of Technology & Engineering	Charotar Institute of Technology	B. Tech M. Tech MTM Ph. D
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B. Pharm M. Pharm MPM PGDCT/PGDPT Ph. D
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	M.B.A PGDM Dual Degree BBA+MBA Ph.D
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications	M.C.A/MCAL M.Sc (IT) Dual Degree BCA+MCA Ph. D
Faculty of Science	P D Patel Institute of Applied Sciences	M.Sc Dual Degree B.Sc+M.Sc

		Ph.D
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy ManikakaTopawala Institute of Nursing Bapubhai Desaibhai Patel Institute of Paramedical Sciences	B.PT M.PT Ph.D B.Sc (Nursing) M.Sc PGDHA BOP PGDMLT GNM Ph.D

The development and growth of the institutes have already led to an investment of over Rs.125 Crores (INR 1250 Million). The future outlay is planned with an estimate of Rs.250 Crores (INR 2500 Million). The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 105 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have **400+** core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, **CHARUSAT** has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which has been the foundation of ECC continues to anchor the functioning of **CHARUSAT**. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, Slovak Academy of Sciences, Slovakia, University of Bielsko-Biala, Poland, ISRO, BARC, Zydus Research Centre, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices, National Education Policy 2020 and emerging requirements. Industrial Visits, Study Tours, Expert Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique **CHARUSAT** pedagogy. The programmes are credit-based and have continuous evaluation as an important feature.

The pedagogy is student-centred, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- The curriculum is contemporary and includes feature necessary to impart students the knowledge and skills necessary for undertaking career in academics,

research, industry and entrepreneurship. Specialized courses also equip students with life-skills.

- Participatory and interactive discussion-based classes with extensive use of ICT tools.
- Sessions by visiting faculty members drawn from leading academic institutions, research organizations and industry.
- Regular weekly seminars and presentations.
- Distinguished lecture series, endowment chair talks, specialized interactive sessions with eminent academicians, researchers and entrepreneurs.
- Practical, field-based projects and assignments.
- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and field visits.
- Dissertation projects
- Fellowships for both undergraduate and post graduate students for exposure to research.
- Final Placement through campus interviews.
- Collaborations with research institutes, academic institutions, industries, industry associations and NOGs
- Effective Mentor-Mentee system
- Personalized counselling by expert psychologists

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills. The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT Welcomes You for A Bright Future



**CHAROTAR UNIVERSITY OF SCIENCE &
TECHNOLOGY
Faculty of Science**

**ACADEMIC REGULATIONS
M. Sc. (Biotechnology/Microbiology/Biochemistry) Programme**

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2022

CHARUSAT

ACADEMIC RULES AND REGULATIONS

[Master's Degree (M Sc) Programme in Biological Sciences]

To ensure uniform system of education duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following are the academic rules and regulations.

1. Nomenclature of the degree

The programme shall be called **Master of Science (M Sc)** with specialization in Biotechnology, Microbiology and Biochemistry.

2. System of Education

- The Semester system of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) at Master's levels.
- Each semester will be at least 90 working day duration.
- Every student requires to do a specified course work in the chosen program of specialization and also to complete project/dissertation/industrial training/specialized training if any.
- Courses have been structured following the concepts of Outcome Based Education (OBE).
- **Medium of instruction will be English**

3. Duration of Programme

Postgraduate programme	(M.Sc.)
Minimum	4 semesters (2 academic years)

Maximum time allowed for completion of M Sc programme shall not be more than 8 semesters.

4. Eligibility for admissions

For the admission to M.Sc., programs in the subjects of Biological Sciences a candidate must have obtained a Degree of Bachelor of Science from any recognized University or a Degree recognized as equivalent thereto.

5. Mode of admissions

Admission to M.Sc. programme will be purely on merit and performance of the candidate at graduation.

6. *Programme structure and Credits*

A student admitted to a program should study the course and earn credits specified in the course structure.

A student admitted to the program should study the prescribed courses and earn credits specified in the course structure. Courses in a programme will be of five kinds: (a) Core, (b) Discipline Specific electives (c) Ability Enhancement Compulsory Courses, (d) Skill Development Course sand (e) Foundation courses. These courses can be further segregated as follows:

- (a) **Core courses (Theory and Laboratory):** These are the courses which should compulsorily be studied by a candidate and are called “Core compulsory”.
- (b) **Discipline Specific elective:** These are the courses where choice or an option for the candidate is available. The student can choose the courses from the sister/allied disciplines which supports the main discipline.
- (c) **Ability Enhancement Compulsory Course (AECC):** These are the courses based upon the content that leads to knowledge enhancement
- (d) **Skill Development Course (SEC):** The are value-based and/or skill-based and are aimed at providing hands-on-training, competencies, skills, etc
- (e) **Foundation courses:** These are the courses based upon the content that leads to knowledge enhancement and/or skill enhancement. These are mandatory for all disciplines but choices are available to the candidate from the defined courses.
- (f) **University Electives:** These are diverse courses of interdisciplinary nature offered by different faculties/centres of the University
- (g) **Additional/Value-added credits:** Review papers, Research Seminars, Community Development Initiatives.
- (h) **Project work/Dissertation:** Special course involving application of knowledge in solving / analyzing /exploring a real-life situation / difficult problem.

Besides, the students are exposed to diverse scientific developments, hands-on skills and entrepreneurial interface through summer internships, scientific seminars/symposia/workshops/expert lectures and similar initiatives

7. *Attendance*

- All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.
- Minimum student attendance required is 80%.

8. *Course Evaluation*

The performance of every student in each course will be evaluated as follows:

- Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and Final examination will be conducted by the University for 70% of the marks for the course.
- Internal evaluation will be based on internal tests and several other tools of assessment like Written tests, MCQ based quizzes, Presentations, Mini-projects, Projects, dissertation, Field visits, industry visits, Seminars, assignments, group discussions/activities, Viva-voce, Online tests etc. as prescribed by the concerned teacher and decided by the Faculty of Science.
- One internal practical test/viva will be conducted per semester totaling to 30 % internal marks for practicals.
- In “Continuous evaluation” Students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the practicals and record keeping.
- The final examination for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or online tests or a combination of any two or more of the above.
- In order to earn the credit in a course a student has to obtain grade other than FF.
- Remedial exams may be conducted after fifteen days of declaration of result of University Winter/Summer Examination. Also, one more June-Supplementary exam shall be conducted in the last week of June (end of Academic year) for the students who are left from odd / even remedial University Exam.

9. Grading

Grading Scheme: Relative grading based on ten point scale will be adopted.

Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

The class awarded to a student in the programme is decided by the final CGPA

- The student’s performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:
 - (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses in the semester
 - (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
 - (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

- In case, a student, of postgraduate programme or postgraduate diploma, gets less than 40% marks in end-semester examination and less than 50% marks overall (combining continuous evaluation and end-semester examination) in a particular course, he / she will not be graded in that course till he / she reappears in said course and obtains specified minimum marks.
- **A student will not be allowed to move to next year if she/he has not cleared all the courses of previous year.**

9. Awards of Degree

- Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- He/She should have earned at least minimum required credits as prescribed in course structure; and He/She should have cleared all internal and external evaluation components in every course; and He/She should have secured a minimum CGPA of 5.0 at the end of the programme; In addition to above, the student has to complete the required formalities as per the regulatory bodies.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.50$
First class	$7.50 > CGPA \geq 6.50$
Second Class	$6.50 > CGPA \geq 5.50$
Pass Class	$5.50 > CGPA \geq 4.00$

11 Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

12. Provision for Appeal

There shall be a provision for Appeal for a candidate who may be dissatisfied with the Grade he/she has been awarded. He/she can approach the University with the written submission.

Annexure-I
Policy for distribution of Internal marks

Particulars	Marks	Marks distribution for 2 credit course	Marks distribution for 3 or 4 credit course
All the internal test/other continuous assessment tool	15 / 30 Marks Sum of average marks obtained by the student from internal test + marks of other assessment components	10 (Should be converted in to 10 at the end of semester)	20 (Should be converted in to 20 at the end of semester)
Assignment Or Seminar	3 /Course	3	6
Attendance	2/Course	2	4
Total		15	30

**Charotar University of Science and Technology
P. D. Patel Institute of Applied Sciences MASTER
OF SCIENCE**

VISION

To become a preeminent national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES

The Postgraduate would be able to

PEO-1: Apply the acquired knowledge in various fields of sciences to plan and execute tasks.

PEO-2: Have scientific aptitude and competency in solving local and global problems.

PEO-3 Possess an ethical approach, while executing research projects and scientific writing.

PEO-4 Ability to follow life-long learning.

PEO-5 Understand the importance of team work and possess leadership and entrepreneurship traits.

Note: POs are mapped with PEOs.

PROGRAM OUTCOMES

The Programme Outcome are as under:

The Post Graduates would be able to

- 1. Knowledge in sciences:** Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.
- 2. Planning Abilities:** Demonstrate effective planning abilities, including time management, resource management, delegation skills, organizational skills, teamwork, And interpersonal skills.
- 3. Problem analysis:** Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.
- 4. Modern tool usage:** Apply appropriate methodologies, methods and procedures, resources to conceptualize the matters.
- 5. Leadership skills:** Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfill tasks with a compassionate attitude.
- 6. Professional Identity:** Establish as a professional in the chosen area of the profession with social responsibility.

7. **Ethics in Science:** To understand the value of being ethical and implement its principles while carrying out research and publication, tasks at the workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/laws related to scientific research and business.
8. **Communication:** Communicate effectively and responsibly with the scientific community and with society at large, such as being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
9. **Modern Science and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.
10. **Environment and sustainability:** Understand the causes of environmental pollution and how it can be reduced and remedies using green technology in industries and environment.
11. **Life-long learning:** Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

Teaching scheme for
(Biotechnology/Microbiology/Biochemistry)

M. Sc. Biotechnology (Semester I)

Course Type	Semester - I							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS721	Microbiology	3	-	-	3	3	30	70	100	-	-	-	100
	MS722	Biochemistry	4	-	-	4	4	30	70	100	-	-	-	100
	MS723	Cell Biology	3	-	-	3	3	30	70	100	-	-	-	100
	MS724	Molecular Biology	4	-	-	4	4	30	70	100	-	-	-	100
	MS725	Evolution and Ecology	2	-	-	2	2	15	35	50				50
Core Course (Laboratory)	BT721	Experimental Skills in Biotechnology-I	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS105.02E	Academic Speaking and Presentation Skills	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS726	Science for Community Development	-	2	-	2	2	-	-	-	-	-	-	-
University Elective	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			16	10		32	26	-	-	-	-	-	-	750

M. Sc. Biotechnology (Semester II)

Course Type	Semester - II							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS771	Bioanalytical Methods and Techniques	3	-	-	3	3	30	70	100	-	-	-	100
	MS772	Molecular Genetics	3	-	-	3	3	30	70	100	-	-	-	100
	MS773	Genetic Engineering	3	-	-	3	3	30	70	100	-	-	-	100
	BT771	Bioprocess Engineering	3	-	-	3	3	30	70	100	-	-	-	100
	BT772	Animal Biotechnology	3	-	-	3	3	30	70	100				100
Core Course (Laboratory)	BT773	Experimental Skills in Biotechnology-II	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS106.02E	Academic Writing	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS774	Review Paper	-	1	-	1	1	-	-	-	-	-	-	-
University Elective	-	University Elective	-	2	-	2	2	-	-	-	30	70	100	100
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			15	11		32	26	-	-	-	-	-	-	900

M. Sc. Microbiology (Semester I)

Course Type	Semester - I							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS721	Microbiology	3	-	-	3	3	30	70	100	-	-	-	100
	MS722	Biochemistry	4	-	-	4	4	30	70	100	-	-	-	100
	MS723	Cell Biology	3	-	-	3	3	30	70	100	-	-	-	100
	MS724	Molecular Biology	4	-	-	4	4	30	70	100	-	-	-	100
	MS725	Evolution and Ecology	2	-	-	2	2	15	35	50				50
Core Course (Laboratory)	MI721	Experimental Skills in Microbiology-I	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS105.02E	Academic Speaking and Presentation Skills	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS726	Science for Community Development	-	2	-	2	2	-	-	-	-	-	-	-
University Elective	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			16	10		32	26	-	-	-	-	-	-	750

M. Sc. Microbiology (Semester II)

Course Type	Semester - II							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS771	Bioanalytical Methods and Techniques	3	-	-	3	3	30	70	100	-	-	-	100
	MS772	Molecular Genetics	3	-	-	3	3	30	70	100	-	-	-	100
	MS773	Genetic Engineering	3	-	-	3	3	30	70	100	-	-	-	100
	MI771	Microbial Enzymes and Physiology	3	-	-	3	3	30	70	100	-	-	-	100
	BT771	Bioprocess Engineering	3	-	-	3	3	30	70	100				100
Core Course (Laboratory)	MI772	Experimental Skills in Microbiology-II	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS106.02E	Academic Writing	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS774	Review Paper	-	1	-	1	1	-	-	-	-	-	-	-
University Elective	-	University Elective	-	2	-	2	2	-	-	-	30	70	100	100
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			15	11		32	26	-	-	-	-	-	-	900

M. Sc. Biochemistry (Semester I)

Course Type	Semester - I							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS721	Microbiology	3	-	-	3	3	30	70	100	-	-	-	100
	MS722	Biochemistry	4	-	-	4	4	30	70	100	-	-	-	100
	MS723	Cell Biology	3	-	-	3	3	30	70	100	-	-	-	100
	MS724	Molecular Biology	4	-	-	4	4	30	70	100	-	-	-	100
	MS725	Evolution and Ecology	2	-	-	2	2	15	35	50				50
Core Course (Laboratory)	BC721	Experimental Skills in Biochemistry-I	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS105.02E	Academic Speaking and Presentation Skills	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS726	Science for Community Development	-	2	-	2	2	-	-	-	-	-	-	-
University Elective	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			16	10		32	26	-	-	-	-	-	-	750

M. Sc. Biochemistry (Semester II)

Course Type	Semester - II							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	MS771	Bioanalytical Methods and Techniques	3	-	-	3	3	30	70	100	-	-	-	100
	MS772	Molecular Genetics	3	-	-	3	3	30	70	100	-	-	-	100
	MS773	Genetic Engineering	3	-	-	3	3	30	70	100	-	-	-	100
	BC771	Enzymology	3	-	-	3	3	30	70	100	-	-	-	100
	BC772	Bioenergetics & Metabolism	3	-	-	3	3	30	70	100				100
Core Course (Laboratory)	BC773	Experimental Skills in Biochemistry-II	-	6	-	12	6	-	-	-	60	140	200	200
Discipline specific Elective (DSE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ability Enhancement Compulsory Course (AECC)	HS106.02E	Academic Writing	-	2	-	2	2	-	-	-	30	70	100	100
Skill Development Course (SEC)	MS774	Review Paper	-	1	-	1	1	-	-	-	-	-	-	-
University Elective	-	University Elective	-	2	-	2	2	-	-	-	30	70	100	100
Additional/Value-added credits*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total			15	11		32	26	-	-	-	-	-	-	900

MS721 MICROBIOLOGY
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100		100	

A. Outline of the course:

Sr No.	Title of the units	Minimum numbers of hours
1.	Introduction to microorganisms and methods of their study	10
2.	Microbial evolution and systematics	11
3.	Introduction to viruses	06
4.	Fungal Diversity	09
5.	Microbial nutrition and growth	09

B. Instructional Methods and Pedagogy

The topics will be discussed in interactive classroom sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem solving sessions will also be conducted by respective faculty members on weekly bases. Course materials will be provided to the students from various primary and secondary sources of information. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

C. Detailed Syllabus

Sr. No.	Title of the units	Minimum numbers of hours
1.	Introduction to microorganisms and methods of their study Introduction to microbiology and its evolution, the nature of the microbial world, general features of diverse groups of prokaryotic and eukaryotic microorganisms, microscopy, stains and staining techniques, sterilization, control of microorganisms, isolation, cultivation and preservation of microorganisms	10
2.	Microbial evolution and systematics: The Evolutionary Process in microbes. Origins of Genetic Diversity, The Evolution of Microbial Genomes, The Dynamic Nature of the <i>Escherichia coli</i> Genome, The Species Concept in Microbiology Theories and methods of microbial systematics: Introduction to taxonomy, nomenclature and systematic Types of classification Taxonomic ranks methods of determining evolutionary relationships, phylogenetic relationships of microorganisms, Phylogenetic trees Evolution of three	11

	domains of life. Salient features of major groups of Domain Prokaryotes and Archaea, Bergey's Manual of Systematic Bacteriology	
3.	Introduction to viruses: Classification and structure of viruses, Bacteriophages, animal viruses and plant viruses, life cycle of viruses (representative of each group); Viroids and Prions. Cultivation of viruses	06
4.	Fungal Diversity General characteristics, diversity, classification and economic importance of cyanobacteria, algae and fungi, growth, salient features of Chytridiomycota, Mastigomycota, Zygomycota, Ascomycota, Basidiomycota and Deuteromycota, identification and general cultivation methods for economically important cyanobacteria, fungi and algae	09
5.	Microbial nutrition and growth Principles of microbial nutrition, nutritional categories of microorganisms, culture media for cultivation of microorganisms, batch, fed-batch and continuous growth, synchronous growth, Mathematical nature and expression of microbial growth, measurement of microbial growth, microbial growth curve, attached growth and biofilms. Influence of environmental factors on microbial growth	09

Total Hours: 45

D. Recommended Study materials:

Text Books:

- Brock Biology of Microorganisms (13th Edition) by Madigan, Martinko, Stahl
- Prescott's, Microbiology, (10th Edition)
- Principles of Microbiology by R.M Atlas

Reference books:

- Bergey's Manual of Systematic Bacteriology : 2nd Edition

E. Course Outcome:

Upon successful completion of the course, student will be able to

- CO1 Comprehend the role of diverse groups of microorganisms in the environment.
- CO2 Recognize, describe, and differentiate between the structure and function of prokaryotic and eukaryotic cells.
- CO3 Describe and analyze the evolutionary process and evolutionary relationship amongst microorganisms
- CO4 Apply knowledge to isolate and classify novel microorganisms from the environment and study its taxonomy.
- CO5 Describe the nutritional requirements of microorganisms, and the pattern of their growth when cultivated in laboratory as pure cultures.

MS722 BIOCHEMISTRY
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100		100	

A. Outline of the course:

Sr. No.	Title of the units	Minimum numbers of hours
1.	Chemical basis of life	12
2.	Biomolecules	20
3.	Metabolism of biomolecules	18
4.	Enzyme & Vitamins	10

B. Instructional Methods and Pedagogy

The topics will be discussed in interactive classroom sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem solving sessions will also be conducted by respective faculty members on weekly bases. Course materials will be provided to the students from various primary and secondary sources of information. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

C. Detailed Syllabus

Sr. No	Title of the units	Minimum numbers of hours
1.	Chemical basis of life ; Structure of atoms, molecules and chemical bonds; Composition of living matter; Water – properties, pH, ionization and hydrophobicity; Emergent properties of biomolecules in water. Stabilizing interactions (Van der Waals, electrostatic, hydrogen bonding, hydrophobic interaction, etc.). Principles of biophysical chemistry (pH, buffer, reaction kinetics, thermodynamics, colligative properties). pH and Buffers: Bronsted-Lowry Concept of Acids and Bases, Buffers: Henderson Hasselbalch equation, Biological buffer systems: The phosphate buffer system, The bicarbonate buffer system, The protein buffer system, The amino acid buffer system, The hemoglobin buffer system.	12
2.	Biomolecules Carbohydrate: Key structures, Monosaccharides, Disaccharides, Polysaccharides, Glycoconjugates, Glycoproteins and Glycolipids,	24

	mutarotation, annomerisation, epimerization, stability of polysaccharides. Introduction to Sugar Code Protein: Introduction to proteins. Basics of protein Structure and function. Primary Secondary, Tertiary and Quaternary Structure of proteins; α-helix, β-sheet and collagen structure helix-coil transition, Ramachandran plots, amino acid sequences. Protein folding. Lipid: Fatty acids, glycerols, phospholipids, sphingolipids, sterols, lipoproteins, membrane phospholipids and prostaglandins; their role in biological membranes. Biological functions of eicosanoids (prostaglandin, leucotrienes and thromboxane). Nucleic acids - structure, diversity and function. Watson-Crick model of DNA; sugar puckerings, stacking; B-; A-and ZDNA ; denaturation kinetics of DNA , Cot curves; structure of tRNA and ribosomes.	
3.	Metabolism of biomolecules Carbohydrate metabolism: Glycolysis, Kreb's cycle, oxidative phosphorylation and Pentose Phosphate Pathway, Gluconeogenesis, glycogenesis, and glycogenosis. Galactose metabolism, Fructose metabolism, aminosugar and mucopolysaccharides metabolism, and their integration. Lipid metabolism: Cholesterol metabolism. Lipid Metabolism: Lipid profile, degradation and biosynthesis and regulation of fatty acids, Transport of fatty acid, oxidation of fatty acids (α, β, ω) Oxidation of unsaturated and odd chain fatty acids; Metabolism and regulation of membrane lipids, Metabolism, regulation and fate of cholesterol.	15
4.	Enzyme & Vitamins Structure and characteristics of enzymes, Enzyme classification, enzyme catalyzed reactions; enzyme inhibition and types of inhibitors. : Enzyme mechanisms and regulation. Classification, structure and physiological roles of Vitamins. Nomenclature, chemistry, digestion, absorption, bioavailability, transport and storage; loss in preparation and handling; functions and mechanisms of action; interaction with other nutrients, deficiency, toxicity and clinical assessment of vitamins: Fat soluble vitamins - A, D, E, K; water soluble- B and C vitamins. Vitamin antagonists in foods.	09

Total Hours: 60

D. Reference Books:

- Lehninger, Nelson and Cox, Principles of Biochemistry, 7th Edition, W.H.Freeman & Company, 2017.
- Fundamentals of biochemistry- Life at the molecular level By: Voet, Voet and Pratt, 5th edition, 2016.
- Lubert Stryer, Biochemistry, 9th Edition, W.H.Freeman and Company, 2019
- Harper's Biochemistry by Robert.KK.Murray 26/e, 2003.

- Fundamentals of Biochemistry by Debajyothi Das 11/e, 2002.
- Human Biochemistry by James. M. Orten & Oho.W. Neuhaus
- Brock biology of microorganism by MT Madigan and JM Mart
- Biochemistry by Jeremy Berg, John Tymoczko and Lubert Stryer, Fifth Edition, Freeman Company;
- Fundamentals of biochemistry- Life at the molecular level By: Voet, Voet and Pratt, Second edition, 2006.
- Human Biochemistry by James. M. Orten & Oho.W. Neuhaus, 10/e, 1983.

E. Course Outcome:

Students will be able to:

- CO1 Comprehend the chemical structures and chemistry of biological polymers (proteins, carbohydrates, lipids and nucleic acids) and their monomers (amino acids, sugars, fatty acids and other lipid monomers, and nucleotides).
- CO2 Elaborate the roles of these biological molecules in living cells.
- CO3 Analyze enzyme kinetic data as well as the bioenergetics/thermodynamics of biochemical reactions.
- CO4 Comprehend the reactions of major metabolic pathways: Central Metabolism (Glycolysis-Gluconeogenesis, Pentose phosphate Pathway, Citric Acid Cycle and Respiratory Electron Transport System, Glyoxylate Cycle; Beta-oxidation of fatty acids and be able to analyze the regulation of these pathways.

MS723 CELL BIOLOGY
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100		100	

A. Outline of the course:

Sr. No.	Title of the units	Minimum numbers of hours
1.	Cell Structures and Biomembranes	8
2.	Intracellular compartments: Organization and Functions	11
3.	Cytoskeleton and Cellular interactions	11
4.	Cell signaling	7
5.	Cell cycle and its regulation	8

B. Instructional Methods and Pedagogy

The topics will be discussed in interactive classroom sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem solving sessions will also be conducted by respective faculty members on weekly bases. Course materials will be provided to the students from various primary and secondary sources of information. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

C. Detailed Syllabus

Sr. No.	Title of the units	Minimum numbers of hours
1	Cell Structures and Biomembranes Overview of cell structures, specialized cells, biomembrane structure and composition, membrane transport mechanisms, electrical properties of membrane	8
2	Intracellular compartments: Organization and Functions Nucleus, endoplasmic reticulum, golgi bodies, lysosomes: protein targeting, protein sorting and protein trafficking; vesicular transport, Mitochondria and chloroplasts, Vacuoles, Peroxisomes, Glyoxysomes	11
3	Cytoskeleton and Cellular interactions Cytoskeleton- Microtubules, microfilaments and intermediate filaments, Extracellular matrix (ECM), Cell-Cell and Cell-ECM interactions, Cell adhesion molecules-cadherins, Immunoglobulin like molecules, integrins and selectins, Cell junctions	11
4	Cell signaling	7

	Overview of modes of intracellular signalling, cell surface receptors, intracellular receptors, signal transduction cascades	
5	Cell cycle and its regulation Cell cycle phases, Regulation of cell cycle- molecular events and model systems, cell death mechanisms-Apoptosis and Necrosis, Implications in Cancer, Basics of Cell Differentiation and Stem Cells	8

Total Hours: 45

D. Recommended Study materials:

- Molecular biology of the Cell by Bruce Alberts.
- Cell and Molecular Biology‘ 8th Edition by E.D.P. De Robertis
- Molecular biology of the Cell by Lodish
- Karp, G. 2010. Cell and Molecular Biology: Concepts and Experiments. 6th Edition. John Wiley & Sons. Inc
- Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. 2009. The World of the cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco
- Essentials of Cell and Molecular Biology: by de Robertis E. D. P. and E. M. F. , Holt Saunder's International Edition (new edition)
- Cell Biology, Genetics, Molecular Biology: Evoloution and Ecology P S Verma and V K Agarwal, S Chand publications, 2004
- Cell Biology 2nd Edition, by Seong S. Han, Ruth Ashley, Gary Hann Rastogi publications
- Essentials of Molecular Biology, 4th edn., by Malacinski GM (2003) Jones & Batielt, London. (ISBN: 0-7637- 2133-6)
- The Physiology and Biochemistry of Prokaryotes, by White, D (2000) Oxford University Press, Oxford
- The Biochemistry of Cell Signalling- Ernst J. M. Helmreich (Indian Edition)-2005 Oxford University Press

E. Course Outcome:

Upon successful completion of the course, student will be able to

- CO1 Recognize, describe, and differentiate between prokaryotes and eukaryotes cells and detailed physiological functions of their cellular components
- CO2 Comprehend the role of cell communication and signaling mechanisms to co-ordinate physiological processes
- CO3 Apply the acquired knowledge to understand and analyze the experimental outcomes in cell biology

MS724 MOLECULAR BIOLOGY
Semester I

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100	-	100	

A. Outline of the syllabus

Sr No	Title of Units	Minimum Number of Hours
1	Organization of genes and chromosomes	12
2	DNA replication, repair and recombination	12
3	RNA synthesis and processing	12
4	Protein synthesis and processing	12
5	Control of gene expression at transcription and translation level	12

B. Instructional Methods and Pedagogy

The topics will be discussed in interactive classroom sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem solving sessions will also be conducted by respective faculty members on weekly bases. Course materials will be provided to the students from various primary and secondary sources of information. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to students in order to improve their performance

C. Detailed Syllabus

No	Modules	Minimum numbers of hours
1	Organization of genes and chromosomes DNA structures and topology, Operon, unique and repetitive DNA, interrupted genes, gene families, structure of chromatin and chromosomes, heterochromatin, euchromatin, transposons	12
2	DNA replication, repair and recombination Unit of replication, enzymes involved, replication origin and replication fork, fidelity of replication for virus, prokaryotes and eukaryotes. Extrachromosomal replicons, telomere and telomerase, DNA damage and repair mechanisms, homologous and site-specific recombination.	12
3	RNA synthesis and processing Transcription factors and machinery, initiation complex formation, transcription activator and repressor, RNA polymerases, capping, elongation, and termination, RNA processing, RNA editing, splicing, and polyadenylation, structure and function of different types of RNA, RNA transport.	12

4	Protein synthesis and processing Ribosome, formation of initiation complex, initiation factors and their regulation, elongation and elongation factors, termination, genetic code, aminoacylation of tRNA, tRNA-identity, aminoacyl tRNA synthetase, and translational proof-reading, translational inhibitors, Post- translational modification of proteins.	12
5	Control of gene expression at transcription and translation level Regulating the expression of phages, viruses, prokaryotic and eukaryotic genes, role of chromatin in gene expression and gene silencing	12

Total Hours: 60

D. Recommended Study materials

- Alberts B, Bray D, Johnson A et al. (1997) Essential Cell Biology. London: Garland Publishing.
- Darwin C (1859) On the Origin of Species. London: Murray.
- Graur D & Li W-H (1999) Fundamentals of Molecular Evolution, 2nd edn. Sunderland, MA: Sinauer Associates.
- Watson JD, Hopkins NH, Roberts JW et al. (1987) Molecular Biology of the Gene, 4th edn. Menlo Park, CA: Benjamin-Cummings.
- Genes VII by B. Lewin Oxford University Press, Cell Press, London (2000)

E. Course Outcomes

Upon successful completion of the course, student will be able to

- CO1: Learn the fundamental molecular processes in a prokaryotic and eukaryotic system
- CO2: Comprehend the difference among two systems
- CO3: Categorize the difference among various molecular mechanisms
- CO4: Develop the insights in gene control mechanisms

MS725 - EVOLUTION AND ECOLOGY
Semester I

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	2	-	2	2
Marks	50	-	50	

A. Outline of the Course

Sr No	Title of Units	Minimum Number of Hours
1	Emergence of evolutionary thoughts	10
2	Evolutionary History and Molecular Evolution Mechanism	10
3	Ecology	10

B. Instructional Methods and Pedagogy

The topics will be discussed in interactive classroom sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem solving sessions will also be conducted by respective faculty members on weekly bases. Course materials will be provided to the students from various primary and secondary sources of information. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to students in order to improve their performance

C. Detailed Syllabus

No	Module	Duration
1	Emergence of evolutionary thoughts Lamarck; Darwin–concepts of variation, adaptation, struggle, fitness and natural selection; migration and random genetic drift; Hardy-Weinberg Law; Adaptive radiation; Isolating mechanisms; Speciation; Allopatricity and Sympatricity; Convergent evolution; Sexual selection; Co-evolution. Origin of cells and unicellular evolution: Origin of basic biological molecules; Abiotic synthesis of organic monomers and polymers; Concept of Oparin and Haldane; Experiment of Miller (1953); The first cell; Evolution of prokaryotes; Origin of eukaryotic cells; Evolution of unicellular	10
2	Evolutionary History and Molecular Evolution Mechanism The evolutionary time scale; Eras, periods and epoch; Major events in the evolutionary time scale; Origins of unicellular and multi cellular organisms; Major groups of plants and animals; Stages in primate evolution including Homo. Concepts of neutral evolution, molecular divergence and molecular clocks; Molecular tools in phylogeny, classification and identification; Protein and nucleotide sequence analysis; origin of new genes and proteins; Gene duplication and divergence	10
3	Ecology Habitat and Niche: Concept of habitat and niche; niche width and overlap; fundamental and realized niche; resource partitioning; character displacement.	10

	Population Ecology: Characteristics of a population; population growth curves; population regulation; life history strategies (r and K selection); concept of metapopulation – demes and dispersal, interdemic extinctions, age structured populations. Species Interactions: Types of interactions, interspecific competition, herbivory, carnivory, pollination, symbiosis.	
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Total Hours: 30

D. Recommended Study materials

- Bonner, J. T. 1988. The Evolution of Complexity. Princeton: Princeton Univ. Press.
- Dawkins, R. 1986. The Blind Watchmaker. New York: Norton.
- Gould, S. J. 1989. Wonderful Life. New York: Norton.
- Dawkins, R. 1982. The Extended Phenotype. New York: W. H. Freeman.
- Williams, G. C. 1966. Adaptation and Natural Selection. Princeton: Princeton Univ. Press (destined to be classic).
- Williams, G. C. 1992. Natural Selection: Domains, Levels and Challenges. New York: Oxford Univ. Press.
- Crow, J. F. 1991. Basic Concepts in Population, Quantitative, and Evolutionary Genetics. New York: W. H. Freeman.
- Falconer, D. S. 1981. Introduction to Quantitative Genetics, second ed. London: Longman.
- Hartl, D. L. And A. G. Clark. 1989. Principles of Population Genetics, second, ed. Sunderland, MA: Sinauer.

E. Course Outcomes

Upon successful completion of the course, student will be able to

- CO1: Learn the basics of theories and patterns of evolution
- CO2: Comprehend the community structures and interrelationships
- CO3: Learn the advanced methods to study the evolution
- CO4: Develop the scientific insights for obtaining ecological data

BT721 EXPERIMENTAL SKILLS IN BIOTECHNOLOGY-I
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	12	12	6
Marks	200		200	

A. Outline of the course:

Sr No.	Title of the units	Minimum numbers of hours
1.	Skill sets in Microbiology	30
2.	Tools and Techniques in Cell Biology	30
3.	Qualitative and Quantitative Analysis of Biomolecules	30
4.	Basic Molecular Biology Skills	30
5.	Ecological Skills	30
6.	Supplementary exercises in Biotechnology	30

B. Instructional Methods and Pedagogy

The theoretical basis and principles of each exercise will be discussed through interactive sessions followed through practical sessions in adequately equipped laboratory. Experiments will be carried out by students individually or in small groups as deemed necessary for the conduct of the experiments. Experimental protocols will be discussed with the students. The students will plan, prepare/obtain reagents and other requirements for conducting experiments. Record preparation as a laboratory journal shall be mandatory for all the students. Students shall be monitored and evaluated regularly as a part of continuous evaluation, and suggestions will be given to the student in order to improve their performance. The laboratory sessions will be supported by virtual labs, video demonstrations, visit to other institutions/ laboratories/industries or field visit as appropriate for developing specific skill sets.

C. Detailed Syllabus

Sr. No.	Title of the units	Minimum numbers of hours
1.	Skill sets in Microbiology Basic techniques in microscopy, stains and staining techniques, sterilization, control of microorganisms, isolation, cultivation and preservation of microorganisms, safety in the microbiological laboratory	30
2.	Tools and Techniques in Cell Biology Comparison of prokaryotic and eukaryotic cells, specialized cells, Molecular organization and functions of biomembranes, cell permeability, mechanisms of transport across membranes, cell viability	30
3.	Qualitative and Quantitative analysis of Biomolecules Biochemical calculations, Qualitative and Quantitative analysis for Carbohydrate, Lipid and Protein, separation and estimation techniques	30
4.	Basic Molecular Biology Skills	30

	Molecular biology reagents, DNA and RNA separation and extraction techniques, Quantification and visualisation methods	
5.	Ecological Skills Exposure to various environments and ecosystem Quantitative and Qualitative analysis of Ecological Data, Measurement of diversity using diversity indices.	30
6.	Supplementary exercises in Biotechnology As per the need of the batch	30

Total Hours: 180

D. Recommended Study materials:

- Bergey's Manual of Systematic Bacteriology : 2nd Edition
- Molecular Cloning: A Laboratory Manual: 3rd Edition
- An introduction to Practical Biochemistry by Plummer 3rd Edition
- Elzinga, C. L., Salzer, D. W., Willoughby, J. W., & Gibbs, J. P. (2001a). *Monitoring Plant and Animal Populations*. Wiley-Blackwell.
- Krebs, C.J. 2008. Ecology: The experimental Analysis of Distribution and Abundance(6thEdition), Benjamin Cummings Publ. 688pgs

E. Course Outcome:

Upon successful completion of the course, student possess

- CO1 knowledge of standard operating procedures and good laboratory practices.
- CO2 the skills to study various prokaryotic and eukaryotic cells with respect to their structures, functions and molecular processes
- CO3 the skills for quantitative and qualitative analysis of biomolecules
- CO4 the basic ability to collect and analyze ecological data
- CO5 scientific writing skills for effective presentation of experimental results
- CO6 the spirit of teamwork and leadership qualities and the ethical practices in scientific experimentation

MI721 EXPERIMENTAL SKILLS IN MICROBIOLOGY-I
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	12	12	6
Marks	200		200	

A. Outline of the course:

Sr No.	Title of the units	Minimum numbers of hours
1.	Skill sets in Microbiology	45
2.	Tools and Techniques in Cell Biology	30
3.	Qualitative and Quantitative Analysis of Biomolecules	30
4.	Basic Molecular Biology Skills	30
5.	Skill set for study of ecosystem	30
6.	Supplementary exercises in Microbiology	15

B. Instructional Methods and Pedagogy

The theoretical basis and principles of each exercise will be discussed through interactive sessions followed through practical sessions in adequately equipped laboratory. Experiments will be carried out by students individually or in small groups as deemed necessary for the conduct of the experiments. Experimental protocols will be discussed with the students. The students will plan, prepare/obtain reagents and other requirements for conducting experiments. Record preparation as a laboratory journal shall be mandatory for all the students. Students shall be monitored and evaluated regularly as a part of continuous evaluation, and suggestions will be given to the student in order to improve their performance. The laboratory sessions will be supported by virtual labs, video demonstrations, visit to other institutions/ laboratories/industries or field visit as appropriate for developing specific skill sets.

C. Detailed Syllabus

Sr. No.	Title of the units	Minimum numbers of hours
1.	Skill sets in Microbiology Safety in the microbiological laboratory, Various methods of sterilization and control of microorganisms Basic techniques for cultivation, growth, observation, and identification of microorganisms.	45
2.	Tools and Techniques in Cell Biology Comparison of prokaryotic and eukaryotic cells, specialized cells, Molecular organization and functions of biomembranes, cell permeability, mechanisms of transport across membranes, cell viability	30
3.	Qualitative and Quantitative analysis of Biomolecules Biochemical calculations, Qualitative and Quantitative analysis for Carbohydrate, Lipid and Protein; Basic techniques for separation of biomolecules.	30
4.	Basic Molecular Biology Skills	30

	Reagents in Molecular biology and its preparation, DNA and RNA separation and extraction techniques. Quantification and visualisation methods	
5.	Skill set for study of ecosystem Exposure to various environments and ecosystem Quantitative and Qualitative analysis of Ecological Data, Measurement of diversity using diversity indices.	30
6.	Supplementary exercises in Microbiology As per the need of the batch	15

Total Hours: 180

D. Recommended Study materials:

- Bergey's Manual of Systematic Bacteriology : 2nd Edition
- Molecular Cloning: A Laboratory Manual: 3rd Edition
- An introduction to Practical Biochemistry by Plummer 3rd Edition
- Elzinga, C. L., Salzer, D. W., Willoughby, J. W., & Gibbs, J. P. (2001a). *Monitoring Plant and Animal Populations*. Wiley-Blackwell.
- Krebs, C.J. 2008. Ecology: The experimental Analysis of Distribution and Abundance(6thEdition), Benjamin Cummings Publ. 688pgs

E. Course Outcome:

Upon successful completion of the course, student possess

- CO1 knowledge of standard operating procedures and good laboratory practices.
- CO2 the skills to study various prokaryotic and eukaryotic cells with respect to their structures, functions and molecular processes
- CO3 the skills for quantitative and qualitative analysis of biomolecules
- CO4 the basic ability to collect and analyze ecological data
- CO5 scientific writing skills for effective presentation of experimental results
- CO6 the spirit of teamwork and leadership qualities and the ethical practices in scientific experimentation

BC721 EXPERIMENTAL SKILLS IN BIOCHEMISTRY-I
Semester I

Credit and hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	12	12	6
Marks	200		200	

A. Outline of the course:

Sr No.	Title of the units	Minimum numbers of hours
1.	Hands-on skills in Microbiology	45
2.	Tools and Techniques in Cell Biology	30
3.	Biomolecules: Qualitative and Quantitative Analysis	30
4.	Basic Molecular Biology Skills	30
5.	Practical Skills in Ecological Study	30
6.	Supplementary exercises in Biochemistry	15

B. Instructional Methods and Pedagogy

The theoretical basis and principles of each exercise will be discussed through interactive sessions followed through practical sessions in adequately equipped laboratory. Experiments will be carried out by students individually or in small groups as deemed necessary for the conduct of the experiments. Experimental protocols will be discussed with the students. The students will plan, prepare/obtain reagents and other requirements for conducting experiments. Record preparation as a laboratory journal shall be mandatory for all the students. Students shall be monitored and evaluated regularly as a part of continuous evaluation, and suggestions will be given to the student in order to improve their performance. The laboratory sessions will be supported by virtual labs, video demonstrations, visit to other institutions/ laboratories/industries or field visit as appropriate for developing specific skill sets.

C. Detailed Syllabus

Sr. No.	Title of the units	Minimum numbers of hours
1.	Hands-on skills in Microbiology Basic microscopic techniques, stains and staining techniques, identification of microbes, sterilization and control of microorganisms, isolation, cultivation and preservation of microorganisms, lab safety during microbial handling	45
2.	Tools and Techniques in Cell Biology Comparison of prokaryotic and eukaryotic cells, specialized cells, Molecular organization and functions of biomembranes, cell permeability, mechanisms of transport across membranes, cell viability	30
3.	Biomolecules: Qualitative and Quantitative Analysis Biochemical calculations, Qualitative and Quantitative analysis for Carbohydrate, Lipid and Protein, separation and estimation methods	30
4.	Basic Molecular Biology Skills	30

	Molecular biology reagents, DNA and RNA separation and extraction techniques, Quantification and visualisation methods	
5.	Practical Skills in Ecological Study Exposure to various environments and ecosystem Quantitative and Qualitative analysis of Ecological Data, Measurement of diversity using diversity indices.	30
6.	Supplementary exercises in Biochemistry As per the need of the batch	15

Total Hours: 180

D. Recommended Study materials:

- Bergey's Manual of Systematic Bacteriology : 2nd Edition
- Molecular Cloning: A Laboratory Manual: 3rd Edition
- An introduction to Practical Biochemistry by Plummer 3rd Edition
- Elzinga, C. L., Salzer, D. W., Willoughby, J. W., & Gibbs, J. P. (2001a). *Monitoring Plant and Animal Populations*. Wiley-Blackwell.
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- CO1 knowledge of standard operating procedures and good laboratory practices.
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- CO4 the basic ability to collect and analyze ecological data
- CO5 scientific writing skills for effective presentation of experimental results
- CO6 the spirit of teamwork and leadership qualities and the ethical practices in scientific experimentation